

Direct interaction of Bcl-2 proteins with tubulin ☆

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Abstract

A direct interaction between tubulin and several pro-apoptotic and anti-apoptotic members of the Bcl-2 family has been demonstrated by effects on the assembly of microtubules from pure rat brain tubulin. Bcl-2, Bid, and Bad inhibit assembly sub-stoichiometrically, whereas peptides from Bak and Bax promote tubulin polymerization at near stoichiometric concentrations. These opposite effects on microtubule assembly are mutually antagonistic. The BH3 homology domains, common to all members of the family, are involved in the interaction with tubulin but do not themselves affect polymerization. Pelleting experiments with paclitaxel-stabilized microtubules show that Bak is associated with the microtubule pellet, whereas Bid remains primarily with the unpolymerized fraction. These interactions require the presence of the anionic C-termini of α - and β -tubulin as they do not occur with tubulin S in which the C-termini have been removed. While in no way ruling out other pathways, such direct associations are the simplest potential regulatory mechanism for apoptosis resulting from disturbances in microtubule or tubulin function.

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Tubulin is an important target for anti-cancer drugs and all of the antimetabolic drugs studied so far have induced apoptosis in a variety of cell types. Initially it was believed that disruption of microtubule integrity by colchicine, vinca alkaloids or nocodazole was responsible for the induction of apoptosis, but agents that hyperstabilize microtubules, such as paclitaxel, also induce apoptosis, hence mitotic block was credited with inducing apoptosis. The only property common to these drugs is their propensity to decrease dynamicity of microtubules [1] and it has been suggested that the increasing stability of the microtubule is involved in apoptosis [2]. How these drug effects lead to apoptosis is, however, not entirely clear; it may involve both cytoplasmic and mitochondria-bound tubulin, hence potentially affecting both the extrinsic and intrinsic

(mitochondria-dependent) pathways [3]. Proposed mechanisms for the action of antimetabolic agents include:

1. Interference with mitosis leading to phosphorylation of Bcl-2 and its congeners [4] thus freeing up Bax, Bak, Bid or other pro-apoptotic members of the Bcl-2 family [5,6]. This, in turn, changes the balance between pro- and anti-apoptotic proteins leading to cell death. However, vinca alkaloids and paclitaxel can release cytochrome *c* from isolated mitochondria [7], thus cytoplasmic contributions may not be necessary in all cases.
2. Non-mitochondrial, direct activation of the caspase cascade, effects on mRNA stability, calcium distribution, etc., [3,8,9].
3. Action on the core apoptotic pathway, the mitochondrial permeability transition pore (VDAC, etc.) via mitochondrial tubulin [10] with release of cytochrome *c*, etc., and subsequent activation of the caspase cascade.

Mitochondria are often associated with microtubules and may move along them. Particularly, good examples

☆ *Abbreviations:* Bcl-2, B-cell lymphoma-2; BH3, Bcl homology domain-3; VDAC, voltage-dependent anion channel; Mes, 2-(*N*-morpholino)ethanesulfonic acid; EGTA, ethylene glycol-bis-(β -aminoethyl ether) *N,N'*-tetraacetic acid; DMSO, dimethyl sulfoxide; SDS, sodium dodecyl sulfate; PVDF, polyvinylidene fluoride.

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of such association can be seen in [11,12]; these show crowding of microtubules around mitochondria and, in some cases, attachment by linker proteins to the outer membrane. Although it is difficult to rule out adventitious adsorption of tubulin during organelle isolation [10,13], careful isolation shows that mitochondria contain varying levels of intrinsic tubulin as dimers or small oligomers but not as microtubules. Bernier-Valentin et al. [14,15] have found mitochondrial tubulin by radio-immunoassay in detergent extracts from carefully washed liver mitochondria amounting to 0.4–0.8% of mitochondrial protein. In addition, added tubulin can bind reversibly to mitochondria with an affinity of $K_D \sim 3.7 \times 10^{-8}$ M and a Hill coefficient near 1.0. Tubulin has been localized to the mitochondrial membrane and shown to be associated with voltage-dependent anion channel (VDAC) by co-immunoprecipitation [10,16].

Mitochondrial tubulin might, therefore, prove to be a promising target for antimetabolic drugs. Nevertheless, a direct interaction between members of the Bcl-2 family of pro- or anti-apoptotic proteins and tubulin has, heretofore, received little attention. We now show that such simple interactions can and do occur in vitro. They influence the assembly properties of tubulin and through such binding may influence the availability of the Bcl-2 proteins.

Materials and methods

Proteins. Rat brain tubulin (>99%) was prepared from frozen brains (−80 °C) by warm/cold cycles of polymerization as described [17] and was stored at 24–25 mg/ml in liquid nitrogen until use. Human recombinant Bcl-2 lacking the C-terminal 21 residues (26 kDa) provided as a fusion protein with maltose binding protein (68 kDa) was from Sigma. A similar preparation but lacking 17 C-terminal residues with a His₁₀-Tag was obtained from Oncogene/EMD Biosciences. C-terminal deleted mouse Bad and recombinant Bcl_{xL} (residues 1–212, 27 kDa) were from Sigma. Bid (amino acids 1–195, 22 kDa) and Bax peptide (16 mer) were from Oncogene. Bak fusion peptide, MW—4404 daltons, RQIKIWFQNRRMKWKKMGQVGRQLAIIGDDINRRY, and its negative control, L78A, were from Calbiochem/EMD Biosciences. BH3 peptides of Bax, Bid, and Bcl-2 were obtained from Abgent, San Diego, CA. All liquid samples were dialyzed against Mes assembly buffer (Mes ab: 0.1 M Mes, pH 6.9, 1 mM EGTA, and 1 mM MgCl₂) using 6–8 kDa tube dialyzers from Novagen, and solid samples were reconstituted in Mes ab. Anti-Bid was from Sigma. Anti-rabbit IgG, peroxidase-linked species-specific whole antibody (from donkey) was from Amersham Pharmacia. Tubulin S was prepared according to Knipping et al. [18].

Polymerization. Microtubule assembly was monitored in 3 mm light path cuvettes at 350 nm, slit 1 nm, at 37 °C in a Peltier-thermostated (Cary300) spectrophotometer for 15 or 20 min. The mixtures in Mes ab contained 1 mM GTP, 10% DMSO, Bcl-2 member, and 24–29 μM tubulin, added to start the reaction. Readings were started 20–30 s after mixing.

Electrophoresis. Twelve micromolar tubulin was incubated in Mes ab at 37 °C with 10% DMSO, 1 mM GTP, with or without 20 μM paclitaxel and with the appropriate Bcl-2 protein for 30 min, and pelleted in an airfuge in a prewarmed rotor for 5 min at 30 psi (100,000g). The supernatant solution was separated and the pellet was reconstituted in an equal volume of the above buffer, and both fractions were electrophoresed at 200 V for ~1 h on 12% NuPage Bis–Tris gels (Invitrogen) using Mes SDS running buffer.

Western blots. Electrophoresed samples were transferred to PVDF membranes by the semi-dry Millipore method at 1.2 mA/cm² for 1 h and blocked overnight at 4 °C with 5% non-fat milk in phosphate-buffered saline with 0.1% Tween 20 (blocking solution). Membranes were incubated at room temperature for 1 h each in blocking solution with anti-Bid, diluted 1:5000, and with anti-rabbit IgG, diluted 1:20,000, with rinses of phosphate-buffered saline plus 0.1% Tween 20 in-between. Detection was by the ECL Plus Western blotting System (Amersham Biosciences).

Results

The interaction between tubulin and members of the Bcl-2 family was assessed by two methods: promotion or inhibition of microtubule assembly and co-pelleting with microtubules. A fusion protein of C-terminal truncated human recombinant Bcl-2 with maltose-binding protein (68 kDa) inhibits the polymerization of pure rat brain tubulin (Fig. 1A). This occurred at mole ratios of Bcl-2/tubulin as low as 0.024 (0.012 corrected for impurities) and affected rate, latent period, and extent of polymerization (measured up to 40 min—data not shown). These data indicate rather high affinity binding and such substoichiometric inhibition suggests poisoning at the growing ends as previously shown for the tubulin/colchicine complex [19]. Similar results were obtained with the C-terminal truncated His₁₀-Tag Bcl-2 protein (data not shown), hence this effect is not a function of the maltose-binding moiety. Bcl_{xL} (minus C-terminus) inhibited polymerization but yielded inconsistent results. This may be electrostatic repulsion due to its lower isoelectric point (4.9 vs 6.4) [20].

Classical Bcl-2 proteins contain four partially conserved so-called homology domains (BH1–4) while certain others contain only BH3 which serves as the primary pro-apoptotic death domain, e.g., Bid. The intact ~27 kDa protein inhibits tubulin polymerization at a somewhat higher mole ratio (=0.109) than Bcl-2, as shown in Fig. 1B. Similarly, Bad, another BH3-only member, was found to inhibit microtubule assembly (Fig. 1C). This occurred at a mole ratio of 0.08, a value not unlike that obtained with Bid and again suggests poisoning of the growing ends. These findings suggest that BH3 is not only the main death domain but may also be required for interaction with tubulin. We, therefore, tested the effect of isolated BH3 peptides of the Bcl-2, Bid, and Bax proteins for their effect on tubulin polymerization. The structure of these domains is as follows:

Bax = TKKLSECLKRIGDELDSDNMEL,

Bid = CIRNIARHALQVGDSMDRSIPP,

Bcl-2 = PPVHLLTLRQAGDDFRSRRYRRC.

All of these BH3 domains failed to influence microtubule assembly at mole ratios (BH3/tubulin) of up to 14.2, 14.1, and 5.2, respectively. However, this does not necessarily rule out their contributions to the interaction with tubulin as is clearly shown in Fig. 2 wherein BH3 peptides of Bcl-2 or Bid, at high mole ratios with respect to Bcl-2, block the inhibitory effect of Bcl-2 on tubulin

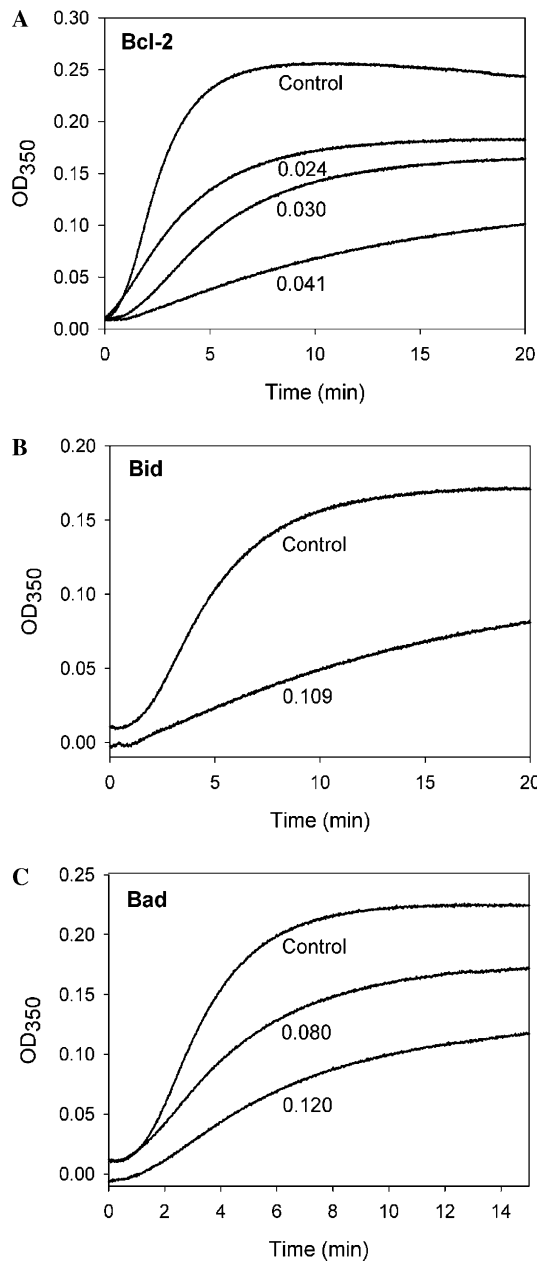


Fig. 1. Inhibition of microtubule assembly by Bcl-2, Bid, and Bad. (A) Bcl-2 maltose-binding fusion protein (0.69 $\mu\text{g}/\mu\text{l}$) in Mes ab, 10% DMSO, and 1 mM GTP was pre-warmed at 37 °C; pure rat brain tubulin (29 μM) was added to start the reaction measured as OD₃₅₀. The numbers on the curves indicate the mole ratios of Bcl-2/tubulin dimer not corrected for impurities in Bcl-2. (B) Dialyzed Bid (0.69 $\mu\text{g}/\mu\text{l}$) was used as in (A). (C) Dialyzed Bad (0.66 $\mu\text{g}/\mu\text{l}$) and 27 μM tubulin as in (A).

polymerization. The Bax BH3 peptide was only weakly effective, if at all (data not shown). Thus, the BH3 domain contributes to the inhibitory effect of the anti-apoptotic members of the Bcl-2 family on microtubule assembly, but additional segments of these proteins are required to affect tubulin polymerization.

Interactions of Bcl-2 proteins with tubulin need not invariably lead to inhibition of polymerization. The 35 amino acid Bak fusion peptide (4.4 kDa) of pro-apoptotic Bak, which binds to and antagonizes Bcl_{XL}, promotes

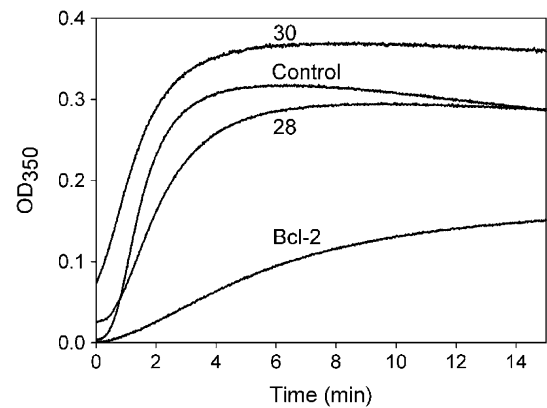


Fig. 2. The effect of BH3 peptides on the response of microtubule assembly to Bcl-2. 2.9 nmol of tubulin was incubated at 37 °C in Mes ab containing 10% DMSO and 1 mM GTP with or without 0.18 nmol Bcl-2 maltose-binding fusion protein (mole ratio = 0.0625) in 100 μl . BH3 peptides of Bcl-2 (11.4 nmol) or Bid (12.3 nmol) were added before tubulin addition. Control, tubulin alone; Bcl-2, tubulin + Bcl-2; 28, mole ratio of BH3(Bcl-2) to Bcl-2; and 30, mole ratio of BH3(Bid) to Bcl-2.

tubulin polymerization at a mole ratio near 1.0 (Fig. 3A). This effect is not a function of Bak's pro-apoptotic properties because a mutant Bak containing an L78A substitution, which is used as a negative control, showed that this peptide also promoted polymerization at a mole ratio near 1.0 (data not shown); therefore, this stimulation appears to be non-specific. As the above two Bak peptides contain an excess of basic amino acids, it was not clear whether or not their assembly-promoting activity was merely charge-based [21]. Hence, we examined the effect of another pro-apoptotic protein, Bax, known to dimerize with Bid or with itself for translocation to the mitochondrion [22]. A Bax peptide composed of residues 150–165 (GWIQDQGGWEGLLSYF), including part of the BH2 homology domain, and containing two acidic and no basic residues, promoted microtubule assembly at mole ratios approaching 1.0 (Fig. 3B). Thus, simple charge-based interaction by the Bcl-2 family members is not a sufficient explanation for the interaction (but see below). It should be noted that the stimulating action of the Bax peptide has some homology with the BH2 domain but not the BH3 domain.

Although members of the Bcl-2 family may either inhibit or promote tubulin polymerization, this effect is not simply related to their pro- or anti-apoptotic activities. Since Bax and Bid or Bcl-2 have opposing effects on tubulin polymerization, Bcl-2 might be expected to mitigate the assembly-promoting effect of Bax. This, in fact, occurs at the respective mole ratios found to be operative when these factors act alone on tubulin (Fig. 3C). Such competition has been pointed out in a number of other situations [23,24].

If Bak and Bax promote tubulin polymerization by binding to tubulin, then they should be associated with microtubules. When assembly was promoted by the addition of paclitaxel, the Bak peptide was found primarily in the microtubule pellet with little remaining in the supernatant solution, as is expected from microtubule assembly promoters (Fig. 4A). The low molecular weight of the

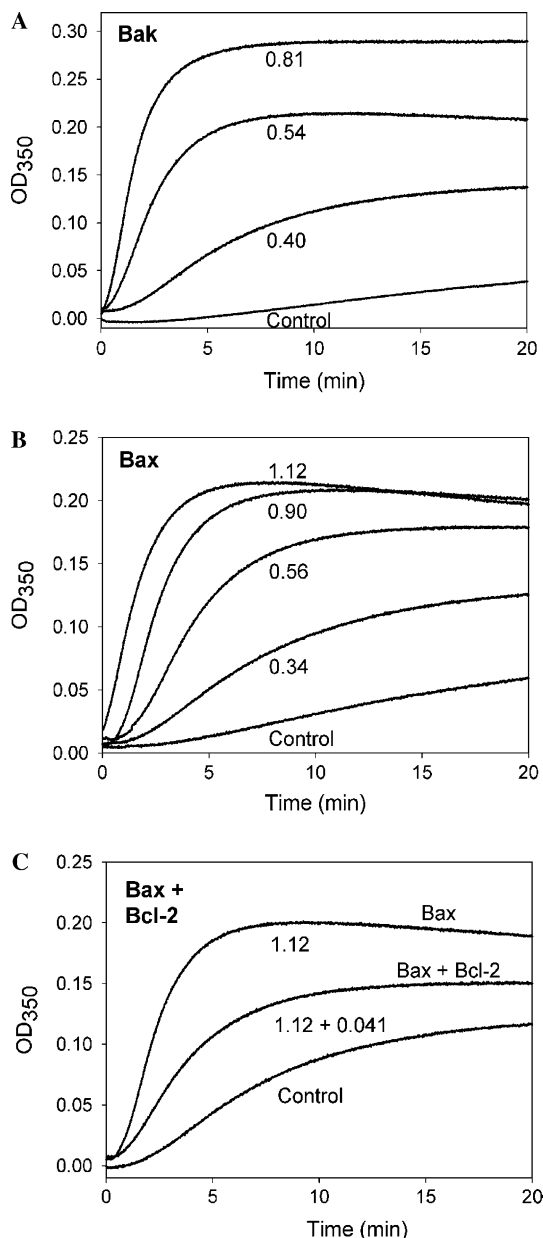


Fig. 3. Stimulation of microtubule assembly by Bak fusion peptide, Bax peptide, or Bax + Bcl-2. (A) Bak peptide (3.27 µg/µl) was dissolved in DMSO, used in Mes ab, 10% DMSO, and 1 mM GTP (final concentrations), and pre-warmed to 37 °C followed by addition of 26 µM pure rat brain tubulin. The numbers on the curves indicate the mole ratios of peptide/tubulin dimer. (B) Bax peptide (0.5 µg/µl) in Mes ab, 10% DMSO, and 1 mM GTP was pre-warmed to 37 °C followed by addition of 26 µM pure rat brain tubulin to start polymerization. Numbers on curves indicate mole ratios of peptide/tubulin dimer. (C) Reversal of Bax assembly stimulation by Bcl-2 with a mole ratio for Bax = 1.12 and for Bcl-2 = 0.041. Conditions as in (B).

Bax peptide prevented SDS gel analysis of such an association. By the same argument, Bid, which prevents polymerization, should be found primarily in the supernatant solution. While this is found to be the case (Fig. 4B), and no Bid was seen in the pellet by Coomassie stain, a small amount of Bid is associated with the microtubule pellet as demonstrated with an antibody (Fig. 4C). A related

‘BH3 only’ protein, Bim, has recently been shown to associate with both dimeric and polymerized tubulin [25] and dynein [26].

Although the charge on the Bcl-2 proteins and peptides may have contributed to the interaction with tubulin, this proved difficult to establish (Fig. 3). However, the charge on tubulin is of great significance for the Bcl-2/tubulin interaction. The C-termini of the α - and β -monomers of tubulin are highly acidic and are involved in the regulation of numerous tubulin functions [27,28]. Their removal by limited proteolysis with subtilisin, to yield tubulin S, leads to profound differences in the properties of tubulin. It was, therefore, important to test whether or not the interaction of Bcl-2 and Bid with tubulin requires the presence of the acidic C-termini. From Fig. 5, it is apparent that the inhibition of tubulin polymerization by His₁₀-Tag Bcl-2 failed to occur with tubulin S, even though the mole ratio of Bcl-2/tubulin was about 6-fold higher. A similar unresponsiveness of tubulin S was observed with the Bcl-2/maltose-binding protein adduct and with Bid (data not shown). Furthermore, the promotion of microtubule assembly by the Bax peptide (Fig. 3) failed to occur with tubulin S (data not shown). Thus, the C-termini are implicated in the interaction of some Bcl-2 proteins with tubulin, suggesting that the acidic C-termini contribute to the interaction. Note that Bcl-2 has a basic ridge on its surface involving particularly adjacent arginines R129, R127, R103, and R12 (as imaged with Rasmol from the NMR coordinates) [20] that may react with the C-termini. A summary of the interactions between Bcl-2 members and tubulin, and the contrasting behavior of tubulin S, is presented in Table 1.

Discussion

Various cytoplasmic events resulting from antimitotic drugs, e.g., Bcl-2 phosphorylation, the accumulation of cell-cycle dependent factors, reduction in microtubule dynamicity, or cell cycle arrest, have been proposed to explain the role of tubulin in apoptosis. However, mitotic arrest per se is not necessarily linked to apoptosis as shown by low doses of antimitotic drugs which block the cell cycle but do not lead to apoptosis [8,29]; moreover, interference with paclitaxel-induced apoptosis by glucocorticoids can occur without an effect on G2/M arrest [3]. Finally, the HIV-1 Tat protein, which causes T Cell apoptosis and releases cytochrome *c* from isolated mitochondria, binds tubulin but causes neither G2/M arrest nor modification of the microtubule network [30]. The fact that drugs binding to the colchicine, vinca, and paclitaxel sites can act directly on isolated mitochondria in the absence of any cytoplasmic contributions [10] indicates that direct attack on tubulin present on or in the outer mitochondrial membrane is another possible locus for disturbing tubulin function. On the assumption that most of these drugs react primarily with tubulin, one must distinguish between an effect of ligand on tubulin and an effect of tubulin on ligand.

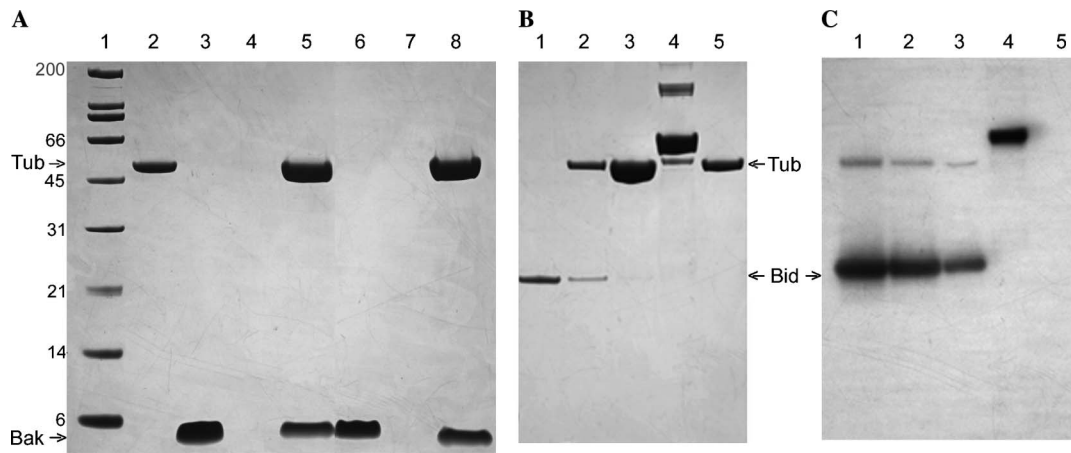


Fig. 4. Binding of Bak and Bid to microtubule pellets. (A) Bak and Bak negative control peptides were each incubated with tubulin under polymerizing conditions and pelleted with centrifugation. Lane 1, molecular weight markers; weights in kDa indicated on the left. Lane 2, tubulin, 3 µg. Lane 3, Bak peptide, 4.4 µg. Lane 4, Bak/tubulin supernatant. Lane 5, Bak/tubulin pellet. Lane 6, Bak negative control peptide, 4.4 µg. Lane 7, Bak negative control/tubulin supernatant. Lane 8, Bak negative control/tubulin pellet. (B) Bid and tubulin prepared as in (A). Lane 1, Bid, 4 µg. Lane 2, Bid/tubulin supernatant. Lane 3, Bid/tubulin pellet. Lane 4, anti-Bid, 1 µl. Lane 5, tubulin, 3 µg. (C) Proteins from (B) were transferred to PVDF membrane and probed with antibody to Bid. Lane 1, Bid. Lane 2, Bid/tubulin supernatant. Lane 3, Bid/tubulin pellet. Lane 4, anti-Bid. Lane 5, tubulin. Note that the heavier bands in lanes 1–3 are not tubulin (no band in lane 5), but are most likely part of the antibody preparation. When the membrane was over-exposed, the band was visible in lane 4. (A,B) NuPage 12% Bis-Tris gels. (C) Western blot of (B).

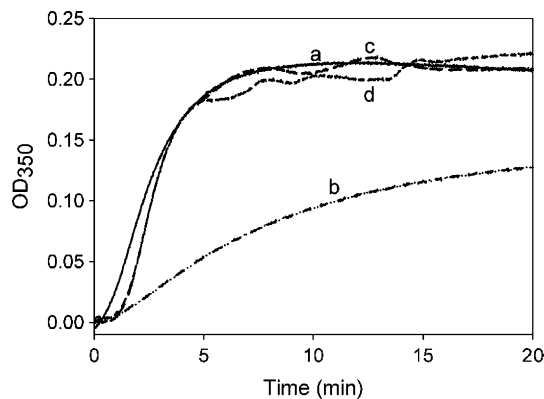


Fig. 5. Comparison of the response of tubulin and tubulin S to Bcl-2 proteins. Concentrations were adjusted to yield approximately the same OD₃₅₀. Twenty-nine micromolar tubulin or 4.6 µM tubulin S was incubated at 37 °C in 10% DMSO with C-terminal truncated His₁₀-Tag Bcl-2 and followed at 350 nm. (a) Tubulin control; (b) tubulin + Bcl-2; (c) tubulin S control; (d) tubulin S + Bcl-2. Mole ratios for tubulin = 0.02 and for tubulin S = 0.127. The oscillations in the tubulin S plateau are repeatedly observed.

Tubulin is known to bind to a large number of proteins and some of these are tumor suppressors, apoptosis-related, and/or mitochondria-associated proteins. They include hexokinase [31]; VDAC but not ANT (adenine nucleotide transporter) [10]; Mmdp1, a linker that binds mitochondria to microtubules in *Saccharomyces pombe* [32]; p53 [2]; the tumor suppressor, Fhit, [33], and APC (adenomatous polyposis coli) [34]. In addition, MAP2 (microtubule-associated protein) is attached to the outer mitochondrial membrane [35] and the Bcl-2 protein Bim is associated with cytoplasmic dynein [26] as a possible reservoir for delivery to the mitochondrion. What might be the functional consequences of such associations? It seems unlikely that the

tubulin embedded in the mitochondrial membrane as dimers or small oligomers [14,15] will be subjected to the same dynamicity or cell cycle arrest factors as are microtubules. On the other hand, direct and indirect evidence for these small molecule tubulin ligands shows that their binding leads to conformational changes in the dimer.

Tubulin ligands change the accessibility of tubulin toward proteases, sulfhydryl reagents, and antibodies [36–44]. By contrast, effects of tubulin on ligands, though less well documented, are found in the altered circular dichroic spectrum of colchicine upon binding to tubulin [45] and in the changes of activity of glycolytic enzyme upon binding to tubulin (e.g. [29]). Because of the cellular excess of tubulin over the Bcl-2 proteins, this mechanism may be the more important one. By either means, conformational changes in the tubulin dimer, or binding-induced changes in its Bcl-2 ligands, could regulate the availability of members of the Bcl-2 family or otherwise influence their effectiveness. Since both pro- and anti-apoptotic proteins bind to tubulin, one must assume that those of lower affinity are more easily released following a conformational change induced by a ligand. Thus, Bcl-2, Bid, and Bad might remain bound while Bax, e.g., would be released thus changing the ratio of *free* pro- to anti-apoptotic proteins. It is of interest that paclitaxel treatment of endothelial cells increased the ratio of Bax/Bcl-2 [46]. We would thus propose that in this simplest of all tubulin mechanisms for apoptosis so far elucidated (though by no means the only one), drugs could release, e.g., Bax from the tubulin present in the outer mitochondrial membrane to bind to components of the permeability transition pore, e.g., VDAC, leading, in turn, to eventual release of cytochrome *c*, etc., and apoptosis. Conversely, the drugs could release tubulin from a binding partner in the mitochondrion, such as VDAC

Table 1
Interaction of Bcl-2 with tubulin

Protein or Peptide	Apoptotic function	Charge or (pI)	Effect on tubulin polymerization	Effect on tubulin S
Bad ^a	Pro-apoptotic	(6.6)	Inhibits	Does not inhibit
Bcl-2	Anti-apoptotic	(6.4)	Inhibits	Does not inhibit
Bcl-xL	Anti-apoptotic	(4.9)	Inhibits	
Bid ^a	Pro-apoptotic	(4.83)	Inhibits	Does not inhibit
Bak peptide	Pro-apoptotic	Basic	Stimulates	Does not stimulate
Bak neg. control		Basic	Stimulates	
Bax peptide	Pro-apoptotic	Acidic	Stimulates	Does not stimulate
Bax BH3		−1	No effect	
Bcl-2 BH3		+3	No effect ^b	
Bid BH3		+2	No effect ^b	

^a BH₃ only.

^b Reverses Bcl-2 inhibition.

[10], so the latter will now be susceptible to the apoptotic action of Bax, etc. The association (or failure thereof) of these proteins with tubulin may decide whether the cell will live or die.

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